MANDATE

Make AI architecture load-bearing.

Turn fragmented GenAI, analytics, data-platform, API, microservice, and automation demand into an enterprise architecture system that accelerates delivery while preserving scientific trust, security, privacy, reliability, cost discipline, and reusable learning.

COMPANY MOMENT AND ROLE PRESSURES

WHY NOW

Thermo Fisher serves science through a broad portfolio of instruments, software, laboratory services, diagnostics, and pharmaceutical services. Digital capabilities must connect heterogeneous workflows without forcing false uniformity.

REAL JOB

Define the technical blueprint, governance, reference patterns, review mechanisms, and practical execution guidance that let squads move faster without accumulating invisible risk or one-off architecture.

01 **Innovation velocity vs enterprise assurance**

02 **Local scientific context vs reusable standards**

03 **Platform optionality vs common control points**

04 **Automation ambition vs human accountability**

05 **Scale economics vs resilient operating ownership**

DECISION ARCHITECTURE

- What changes and where attention goes
- What gets funded, reused, partnered or stopped
- What becomes an enterprise standard and what remains a legitimate exception
- What escalates and which human roles retain authority
- What production ownership, evidence and economics are required before scale

OPERATING MODEL

The Scientific Capability Lattice

A reference architecture is load-bearing only when five layers stay connected. Weakness in any one layer should change the disposition of the whole capability.

01 CONTEXT & DATA AUTHORITY

Authoritative sources, lineage, quality, access, semantics, retention and domain ownership.

02 MODELS, AGENTS & INTELLIGENCE

RAG, models, agents, orchestration, evaluation, grounding, abstention and change control.

03 WORKFLOW & INTEGRATION

APIs, microservices, events, ERP/CRM, laboratory systems, RPA, exceptions and human handoffs.

04 TRUST, GOVERNANCE & CONTROL

Privacy, security, compliance, human authority, threat modeling, review evidence and decision records.

05 OPERATIONS, ECONOMICS & LEARNING

SLOs, traces, incidents, support ownership, unit cost, adoption, outcome evidence and pattern reuse.

DISPOSITION LOGIC

PROMOTE when outcome, authority, controls, reliability, economics, adoption, and ownership are evidenced. REVISE when the pattern is useful but one load-bearing condition is weak. HOLD when unresolved risk or ownership would make scale irresponsible. RETIRE when the capability no longer earns its operating burden.

BALANCED SCORECARD

Outcome and cycle time

Pattern reuse and duplication avoided

SLOs, incidents and recovery

Grounding, access, privacy and compliance

Unit cost and realized value

Retained use and exception behavior

Feedback-to-change time

PROOF MAP AND STRONGEST OBJECTION

TRANSFERABLE EVIDENCE

Vape-Jet: current AI-first operations across robotics, telemetry, ERP, CRM, quality and support. DudeWorth: AI readiness, agents, RAG, human judgment, evaluation and governance. Compunetics: high-reliability engineering, controls and complex technical programs. Amazon: scaled operating mechanisms and adoption.

OBJECTION, STATED PLAINLY

Russell has not held the formal Enterprise AI Architect title and does not claim production ownership of Thermo Fisher exact platform estate. The response is not title inflation: pair with specialists, read the current architecture, and prove judgment through one bounded review and reference pattern.

EXECUTIVE QUESTIONS

- Where is architecture friction currently slowing squads most?
- Which AI/data patterns are already trusted and should be protected?
- Where do local scientific requirements legitimately override enterprise defaults?
- Who owns production reliability, model/knowledge change, and incident response today?
- What evidence is required to promote a pattern from local proof to enterprise reference?
- How are cost, adoption, and business outcomes connected to architecture decisions?

FIRST 120 DAYS

Read the system -> choose one bounded proof -> run one reference pattern -> compound the learning into reusable assets, clear decision rights, and an evidence-based disposition cadence.

CHANGE BURDEN

- Business and scientific owners name workflow outcomes, authoritative context and acceptable risk.
- Platform and architecture teams expose paved roads, failure boundaries, cost and operational ownership.
- Security, privacy, compliance and data governance participate early enough to shape design, not merely approve it.
- Squads capture exceptions, adoption friction and evidence so patterns can learn.

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